

BRENDAN HARRIS

Computational neuroscientist

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RESEARCH THEMES

I study how collective dynamics in neural populations perform computation, combining large-scale electrophysiology with statistical physics and dynamical systems theory. My methods rely on uncovering structure from noisy, high-dimensional, heavily subsampled recordings.

- **Criticality and computation:** tracking distance to a critical point when the noise process is unknown and sampling is sparse.
- **Spatiotemporal dynamics:** nested traveling waves that coordinate hierarchical processing and communication between brain areas.
- **Stochastic processes:** fractional diffusion for spiking-circuit mean field theory, decision making, and adaptive visual sampling.
- **Population geometry:** coding and communication subspaces, dimensionality reduction for large-scale electrophysiology data.

TECHNICAL SKILLS

Neural data: high-density extracellular electrophysiology (Neuropixels, multi-electrode arrays); spike trains, LFP, and EEG; spectral, phase, and traveling-wave analysis; criticality and phase-transition analysis.

Statistics and modeling: dimensionality reduction; stochastic processes and statistical physics; spiking neural networks; mean-field reductions; feature-based time-series analysis (*hctsa*, *catch22*) and machine-learning.

Engineering: Julia, Python, MATLAB, C++; real-time and closed-loop systems; hardware interfacing; HPC and large-scale data pipelines; reproducible analysis, package release, and long-term maintenance.

INDUSTRY EXPERIENCE

2023

Scientific software developer, real-time closed-loop neurofeedback

Developed real-time software for Resonait’s EEG neurofeedback platform, targeted at monitoring and treating depression; a commercial product built by a small team.

- Interfaced with EEG headset hardware and built the online acquisition and signal-processing pipeline.
- Analyzed large-scale EEG datasets with online hidden Markov models.
- Wrote technical documentation and managed the codebase.
- Co-authored peer-reviewed paper, “Default mode network detection using EEG in real-time” (IEEE EMBC 2024).

Resonait Medical Technologies

SELECTED PUBLICATIONS

6 peer-reviewed publications, two as first author in Q1 journals (*Physical Review X* and *Nature Communications*, top 3%). All first-author work is accompanied by [open-source code](#).

2026

Nested spatiotemporal theta–gamma waves organize hierarchical processing across the mouse visual cortex

Harris, B. & Gong, P.

New analytical methods for high-density multi-electrode recordings.

Nature Communications

2025

Canonical time-series features for characterizing biologically informative dynamical patterns in fMRI

Alam, I., Harris, B., Cahill, P., Cliff, O., Markicevic, M., Zerbi, V. & Fulcher, B. D.

Analyzed mouse cell densities and fMRI recordings.

Aperture Neuro

2024

Tracking the distance to criticality in systems with unknown noise

Harris, B., Gollo, L. L. & Fulcher, B. D.

New criticality measure robust to unknown noise and sparse sampling.

Physical Review X

2024

Default mode network detection using EEG in real-time

Cooray, N., Gohil, C., Harris, B., Frost, S. & Higgins, C.

Real-time HMM inference and hardware interfacing for closed-loop feedback.

IEEE EMBC

2024

Distributed and dynamical communication: a mechanism for flexible cortico-cortical interactions and its functional roles in visual attention

Ni, S., Harris, B. & Gong, P.

Communication subspace analysis for large-scale electrophysiology.

Communications Biology

EDUCATION

2022–2026	Physics PhD “Cross-scale dynamics in the working regime of the visual cortex”. <i>Supervisor</i> : Pulin Gong. Two chapters published in Q1 journals; third is in preparation (slides).	The University of Sydney
2021	Physics Honours, Class I and the University Medal “Inferring parametric variation across non-stationary time series”. <i>Supervisor</i> : Ben Fulcher.	The University of Sydney
2018–2020	Bachelor of Science / Bachelor of Advanced Studies (Dalyell Scholar) Majored in Physics and Neuroscience.	The University of Sydney

RESEARCH LEADERSHIP AND SCIENTIFIC SERVICE

2026–	Co-organizer, Systems Neuroscience and Complexity seminar series Convene a cross-disciplinary seminar series spanning physics, medicine, and computer science.	The University of Sydney
2025–	Co-supervised Honours research - Jude Metcalf (2026), <i>Adaptive fractional dynamics of visual sampling</i> - Aiden Sloots (2025), <i>Bi-fractional diffusion for decision-making dynamics</i>	The University of Sydney
2018–	Author and maintainer of open-source scientific software AllenNeuropixels.jl (Allen Institute Neuropixels datasets), Catch22.jl , TimeseriesFeatures.jl and TimeseriesTools.jl (feature-based time-series analysis), SpatiotemporalMotifs.jl (wave analysis of electrophysiology data), and Dewdrop.jl (GPU spiking neural circuits).	GitHub: brendanjohnharris
2018–	Public communication Authored an article in The Conversation with a video summary; ongoing science communication via Bluesky, GitHub, and a personal website.	Sydney

SELECTED TALKS, PRESENTATIONS, AND WORKSHOPS

2026	Sydney–Tokyo: Neural and Nonlinear Dynamics Workshop	Talk
2026	Maths in the Brain, Melbourne	Talk (best ECR talk prize)
2025	Center for Neuroscience Imaging Research (CNIR), IBS, South Korea	Invited talk
2025	OHBM 2025, Brisbane; COSYNE 2025, Montreal	Posters
2025	EPC/APCV 2025, UNSW; NeuroEng 2025, The University of Melbourne	Talks
2025	Pre-COSYNE BrainHack 2025, Mila, Montreal	Workshop
2023	Hosted visits: MIP:Lab Geneva; Univ. Melbourne; Monash Univ.	Talks
2022	Yale Medical School (host: Dr. E. Lake)	Invited talk
2022	31st Annual Computational Neuroscience Meeting (CNS*2022), Melbourne	Poster & workshop talk
2022	OpenScope NeuroDataReHack (Allen Institute, Seattle); BrainHack Australia	Workshops

SELECTED AWARDS

2026	Best ECR talk prize, Maths in the Brain 2026	Prize
2025	Presenters travel grant, COSYNE 2025	Award
2022	Research Training Program (fee offset and stipend)	Scholarship
2022	University Medal (Physics); Dean’s List of Excellence	Recognition

REFEREES

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